

1.

4.2.4 (2015:01)

In which of the following compounds is the oxidation number of manganese **lowest**?

- (a) Mn_2O_3
- (b) K_2MnO_4
- (c) NaMnO_4
- (d) MnO_2

2.

4.2.5 (2015:05)

Which one of the following reactions can occur spontaneously at $25.0\text{ }^\circ\text{C}$? (Assume the solutions have a concentration of 1.00 mol L^{-1} .)

- (a) $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$
- (b) $2\text{Fe}^{2+}\text{(aq)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{Fe}^{3+}\text{(aq)} + 2\text{Cl}^-\text{(aq)}$
- (c) $\text{Zn}^{2+}\text{(aq)} + \text{Cu(s)} \rightarrow \text{Zn(s)} + \text{Cu}^{2+}\text{(aq)}$
- (d) $\text{Cu(s)} + 2\text{H}^+\text{(aq)} \rightarrow \text{Cu}^{2+}\text{(aq)} + \text{H}_2\text{(g)}$

3.

4.2.5 (2015:08)

In which one of the following situations will there be no visible reaction?

- (a) Solutions of sodium fluoride and potassium chloride are mixed together.
- (b) A clean strip of copper metal is placed into a silver nitrate solution.
- (c) Bromine water and ethene are shaken together.
- (d) Sodium metal is dropped into a beaker of distilled water.

4.

4.2.5 (2015:10)

When a grey solid was added to a green solution, the **most** obvious observation was decolourising of the solution.

Which one of the following might have been the reactants?

	Solid	Solution
(a)	chromium	lead(II) nitrate
(b)	zinc	cobalt(II) nitrate
(c)	cadmium	nickel(II) nitrate
(d)	lead	chromium(III) nitrate

5.

4.2.8 (2015:17)

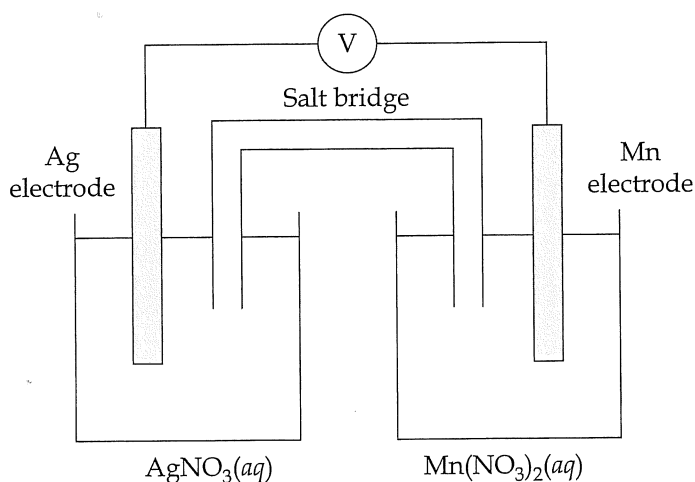
A half-cell containing a metal electrode in a sodium nitrate solution is joined to another half-cell containing an inert electrode in a metal nitrate solution. Which one of the following combinations of electrode and metal nitrate solution will produce an electrochemical cell with the greatest electrical potential under standard conditions?

	Electrode	Nitrate solution
(a)	Ag	Sn^{2+}
(b)	Al	Cd^{2+}
(c)	Cr	Fe^{2+}
(d)	Cu	Fe^{3+}

6.

4.2.8 (2016 SP:15)

A galvanic cell consisting of $\text{Ag}|\text{Ag}^+$ and $\text{Mn}|\text{Mn}^{2+}$ couples is constructed as represented by the diagram below. All solutions are 1.0 mol L^{-1} and the temperature is 25°C .



Which one of the following gives the predicted emf for this cell in volts?

- (a) 0.42 (c) 1.98
(b) 0.80 (d) 2.78

7.

4.1 (2016 SP:16)

Consider the following statements about fuel cells.

- (i) A fuel cell is a device that converts chemical energy to electrical energy via a redox reaction.
- (ii) Fuel cell technology involves the continuous supply of reactants to the cells and the continuous removal of the products.
- (iii) A fuel cell can be recharged by reversing the direction of current flow through the cell.
- (iv) Fuel cells are considered a low-emission technology.

Which of the above statements about fuel cells are true?

- (a) i only (c) i, iii and iv
(b) i and ii (d) i, ii and iv

8.

4.2.4 (2016:13)

Nitric acid can be manufactured from the element nitrogen using the steps below.

Step	1	2	3	4
	$\text{N}_2 \rightarrow \text{NH}_3$	$\text{NH}_3 \rightarrow \text{NO}$	$\text{NO} \rightarrow \text{NO}_2$	$\text{NO}_2 \rightarrow \text{HNO}_3$

The smallest change in the oxidation number of nitrogen is found in

- (a) step 1. (c) step 3.
(b) step 2. (d) step 4.

9.

4.2.4 (2016:14)

Which one of the following is **not** a redox reaction?

- (a) $2 \text{CrO}_4^{2-} + 2 \text{H}^+ \rightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$
(b) $2 \text{Cr}^{2+} + 2 \text{H}^+ \rightarrow 2 \text{Cr}^{3+} + \text{H}_2$
(c) $(\text{NH}_4)_2 \text{Cr}_2\text{O}_7 \rightarrow \text{N}_2 + 4 \text{H}_2\text{O} + \text{Cr}_2\text{O}_3$
(d) $\text{Cr}_2\text{O}_3 + 3 \text{C} \rightarrow 2 \text{Cr} + 3 \text{CO}$

10.

4.2.5 (2016:15)

Which one of the following shows the substances listed in order of increasing strength as reducing agents?

- (a) F^- , Al , Zn , Cu , I^-
(b) I^- , F^- , Zn , Al , Cu
(c) F^- , I^- , Cu , Zn , Al
(d) Zn , Al , Cu , I^- , F^-

11.

4.2.7 (2016:16)

Which one of the following reactions would **not** produce a current at 25.0 °C, when set up as a galvanic cell?

- (a) $\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 \text{Fe}^{2+}(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\ell) + 5 \text{Fe}^{3+}(\text{aq})$
(b) $\text{Fe}(\text{s}) + \text{Pb}^{2+}(\text{aq}) \rightarrow \text{Fe}^{2+}(\text{aq}) + \text{Pb}(\text{s})$
(c) $\text{Br}_2(\ell) + 2 \text{Cl}^-(\text{aq}) \rightarrow 2 \text{Br}^-(\text{aq}) + \text{Cl}_2(\text{g})$
(d) $\text{Fe}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Fe}^{2+}(\text{aq}) + \text{Cu}(\text{s})$

4.2.5 (2016:17)

12.

Three metals, A, B and C, were tested to compare their reactivity. Samples of each metal were placed separately into test tubes each containing a nitrate solution of the other metal ions. The following results were obtained.

	A(s)	B(s)	C(s)
$A^{2+}(aq)$		No visible reaction	Solid A forms
$B^{2+}(aq)$	Solid B forms		Solid B forms
$C^{2+}(aq)$	No visible reaction	No visible reaction	

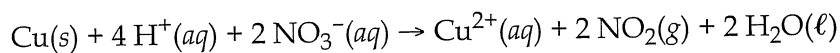
From these results, the metals arranged in order of decreasing strength as reducing agents can be concluded to be

- (a) $C > A > B$.
- (b) $B > C > A$.
- (c) $B > A > C$.
- (d) $A > C > B$.

13.

4.2.4 (2017:02)

Copper reacts with nitric acid as shown in the redox equation below.



Which one of the following states the change in the oxidation number of nitrogen?

- (a) 3+ to 0
- (b) 5+ to 4+
- (c) 3+ to 2+
- (d) 5+ to 0

14.

4.2.6 (2017:03)

In the electrolysis of molten calcium bromide, one mole of bromine molecules is formed for every one mole of calcium. This is because

- (a) the formula of calcium bromide is Ca_2Br .
- (b) the valency on a calcium ion is twice that on a bromide ion.
- (c) bromine is more reactive than calcium.
- (d) the atomic mass of bromine is twice that of calcium.

4.2.6 (2017:04)

15.

Which of the following are common to both galvanic and electrolytic cells?

- (i) a salt bridge
 - (ii) an external circuit
 - (iii) the transfer of electrons and movement of ions
 - (iv) at least two different reactions with distinct reduction potentials
- (a) i and ii only
 - (b) i, ii and iv only
 - (c) ii, iii and iv only
 - (d) i, ii, iii and iv

16.

3.3 (2017:09)

An example of an undesirable electrochemical process is the oxidation of metals. Which one of the following equations does **not** represent what might occur during oxidation?

- (a) $4 \text{Fe(OH)}_2(\text{s}) + 2 \text{H}_2\text{O}(\text{aq}) + \text{O}_2(\text{g}) \rightarrow 4 \text{Fe(OH)}_3(\text{s})$
- (b) $\text{O}_2(\text{g}) + 2 \text{H}_2\text{O}(\ell) + 4 \text{e}^- \rightarrow 4 \text{OH}^-(\text{aq})$
- (c) $\text{FeCl}_3(\text{s}) \rightarrow \text{Fe}^{3+}(\text{aq}) + 3 \text{Cl}^-(\text{aq})$
- (d) $\text{Pb}^{2+}(\text{aq}) \rightarrow \text{Pb}^{4+}(\text{aq}) + 2 \text{e}^-$

17.

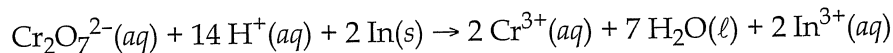
4.2.7 (2017:10)

Which one of the following reactions will be spontaneous under standard conditions?

- (a) $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 3 \text{H}_2\text{O}_2(\text{aq}) + 8 \text{H}^+(\text{aq}) \rightarrow 2 \text{Cr}^{3+}(\text{aq}) + 7 \text{H}_2\text{O}(\ell) + 3 \text{O}_2(\text{g})$
- (b) $3 \text{O}_2(\text{g}) + 4 \text{Au}(\text{s}) + 12 \text{H}^+(\text{aq}) \rightarrow 4 \text{Au}^{3+}(\text{aq}) + 6 \text{H}_2\text{O}(\ell)$
- (c) $2 \text{Ag}^+(\text{aq}) + 2 \text{Br}^-(\text{aq}) \rightarrow 2 \text{Ag}(\text{s}) + \text{Br}_2(\ell)$
- (d) $2 \text{Cl}^-(\text{aq}) + \text{I}_2(\text{s}) \rightarrow \text{Cl}_2(\text{g}) + 2 \text{I}^-(\text{aq})$

Questions (2018:05) and (2018:06) refer to the following information.

When a piece of indium metal, $\text{In}(s)$, is placed in some acidified dichromate solution, $\text{Cr}_2\text{O}_7^{2-}(aq)$, a reaction occurs resulting in $\text{In}^{3+}(aq)$ ions being produced. The equation for this reaction is shown below.



The EMF for this reaction at 25.0°C was found to be $+1.70 \text{ V}$.

18.

4.2.8 (2018:05)

What is the calculated E° value for the In^{3+}/In half-equation?

(a) -0.34 V

(b) 0.34 V

(c) 1.36 V

(d) 3.06 V

19.

4.2.8 (2018:06)

According to the Standard Reduction Potential Table, which of the following sets of metals **cannot** be oxidised by indium ion, In^{3+} , under standard conditions?

(a) Sn , Cd , Fe , Cr

(b) Mg , Na , Ca , Sr

(c) Mn , Ni , Sn , Cu

(d) Ni , Sn , Cu , Ag

20.

4.2.4 (2018:07)

Molybdenum, Mo , is present in each of the following species: MoO_2 , $\text{Mo}_2\text{O}_7^{2-}$, HMoO_4^{2-} .

Which of the following lists these species in order of **increasing** oxidation number of molybdenum?

(a) HMoO_4^{2-} MoO_2 $\text{Mo}_2\text{O}_7^{2-}$

(b) $\text{Mo}_2\text{O}_7^{2-}$ HMoO_4^{2-} MoO_2

(c) $\text{Mo}_2\text{O}_7^{2-}$ MoO_2 HMoO_4^{2-}

(d) MoO_2 HMoO_4^{2-} $\text{Mo}_2\text{O}_7^{2-}$

Questions (2018:14) and (2018:15) refer to the following information.

A group of students decided to investigate the reactivity of four different transition metals; rhenium, vanadium, zirconium and tantalum. They did this by placing small pieces of each metal in separate test tubes with the appropriate test solutions. The 1.00 mol L^{-1} test solutions were prepared by dissolving the nitrate salt of each metal in distilled water. Their results are summarised in the table below.

Metal	Metal Ions			
	$\text{Re}^{3+}(\text{aq})$	$\text{V}^{2+}(\text{aq})$	$\text{Zr}^{4+}(\text{aq})$	$\text{Ta}^{3+}(\text{aq})$
Rhenium		no reaction	no reaction	no reaction
Vanadium	reaction occurs		no reaction	reaction occurs
Zirconium	reaction occurs	reaction occurs		reaction occurs
Tantalum	reaction occurs	no reaction	no reaction	

21.

4.2.3 (2018:14)

Which of these metals is the **most** easily oxidised?

- (a) rhenium (c) zirconium
(b) vanadium (d) tantalum

22.

4.2.3 (2018:15)

Which of these metals is the **weakest** reducing agent?

- (a) rhenium (c) zirconium
(b) vanadium (d) tantalum

23.

4.2.6 (2018:25)

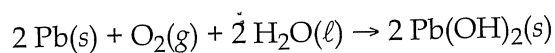
Which of the following statement pairs can be used to distinguish between an electrolytic cell and a galvanic cell?

	Electrolytic cell	Galvanic cell
(a)	an electric current flows from an external electrical power source	the chemical reaction produces an electric current
(b)	oxidation occurs at the cathode	oxidation occurs at the anode
(c)	ions do not migrate through an electrolyte	ions migrate through an electrolyte
(d)	can be used to power a battery	can be used to electroplate metals such as copper and silver

4.2.8 (2019:06)

24.

The overall equation for the reaction of lead with oxygen is as follows:



What is the theoretical E^0 value for the overall Pb/O₂ reaction under standard conditions?

- (a) -0.27 V
- (b) +0.27 V
- (c) +0.53 V
- (d) +0.93 V

4.2.3 (2019:10)

25.

Which statement is correct?

- (a) Fluorine can be oxidised by potassium bromide solution but not by potassium iodide solution.
- (b) Chlorine can be oxidised by potassium fluoride solution but not by potassium iodide solution.
- (c) Chlorine can be reduced by potassium bromide solution but not by potassium iodide solution.
- (d) Bromine can be reduced by potassium iodide solution but not by potassium chloride solution.

4.2.4 (2019:17)

26.

In which of the following sets do all the **bolded** and underlined atoms have the same oxidation number?

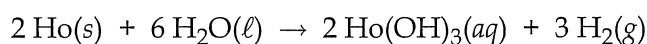
- (i) H₂O, O₂, H₂O₂
- (ii) H₂O₂, NaCl, MgH₂
- (iii) NaCl, Li₂CO₃, KOH
- (iv) FeO, Fe₂O₃, Fe

- (a) i and iv only
- (b) ii and iii only
- (c) iv only
- (d) i, ii and iii only

27.

4.2.4 (2020:01)

Holmium (Ho) reacts quickly with hot water to form holmium hydroxide and hydrogen:



The oxidising and reducing agents in this equation are

	Oxidising agent	Reducing agent
(a)	H ₂ O	H ₂
(b)	Ho	H ₂ O
(c)	H ₂ O	Ho
(d)	Ho(OH) ₃	Ho

28.

4.2.1 (2020:03)

Oxidation–reduction reactions involve the transfer of

- (a) protons.
- (b) electrons.
- (c) hydroxide ions.
- (d) hydrogen ions.

29.

4.2.6 (2020:06)

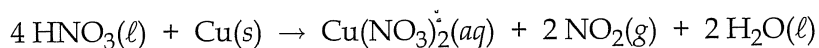
What type of redox reaction occurs in a galvanic cell and what is one possible use for such a cell?

	Type of redox reaction	Possible use of a galvanic cell
(a)	non-spontaneous	the plating of cheap metallic objects with precious metals
(b)	spontaneous	the plating of cheap metallic objects with precious metals
(c)	non-spontaneous	the production of an electric current for a torch
(d)	spontaneous	the production of an electric current for a torch

30.

4.2.2 (2020:07)

The following equation shows the reaction between copper and concentrated nitric acid:



Observable changes associated with this reaction are the dissolving of the copper, the formation of a deep blue solution and the evolution of a pungent brown gas.

Which of the following are some of the atomic/molecular scale events needed for these observable changes to occur?

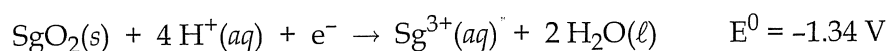
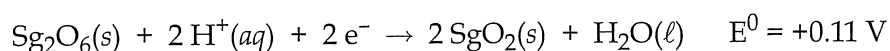
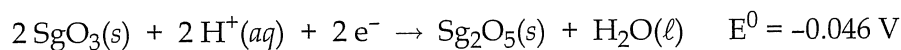
- (i) collisions between HNO_3 molecules and Cu atoms
- (ii) donation and acceptance of protons
- (iii) reduction of copper atoms

- (a) i only
- (b) ii only
- (c) i and iii only
- (d) i, ii and iii

31.

4.2.5 (2020:19)

The following half-equations show some predicted standard reduction potentials for seaborgium (Sg) oxides:



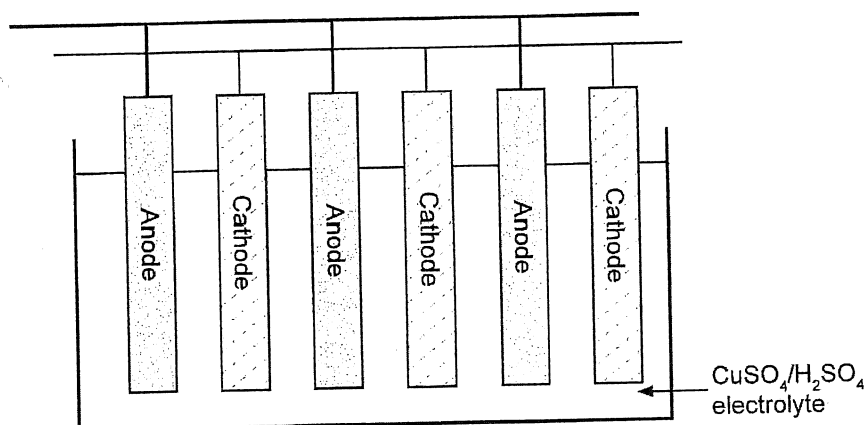
The strongest reducing agent is

- (a) SgO_3
- (b) Sg_2O_5
- (c) SgO_2
- (d) Sg^{3+}

4.2.13 (2020:20)

32.

Impure copper must be purified before it is used in applications where very high electrical conductivity is required. The purification of copper, which is also known as electrorefining, can be performed in an electrochemical cell similar to the one shown below.



Which statement regarding this electrochemical cell is correct?

- (a) This cell requires the application of an external electrical potential difference for it to function.
- (b) During operation, the electrolyte becomes more blue because the concentration of Cu^{2+} ions in the electrolyte increases.
- (c) This cell will not work because it does not have a salt bridge.
- (d) The impure copper is cast as cathodes.

- 5.(2015:21-c) Answer 'c'. The others can be considered as outliers.
- 6.(2016 SP:04-d) The Brønsted-Lowry model is about accepting and losing protons. Answer 'd' is an acid/carbonate reaction. The answer is 'd'.
- 7.(2016 SP:10-a) Original pH was 8.2 = $[H^+]$ of 6.31×10^{-9} and the new $[H^+] = 7.94 \times 10^{-9}$ so there was a 20% change. The answer is 'a'.
- 8.(2016:07-b) Leaving the funnel in place will add drops after the initial reading, having many people making readings will add random errors, rinsing with distilled water will dilute the titrant but measuring from the bottom of the meniscus for both the initial and final reading is not an error so the answer is 'b'.
- 9.(2016:09-a) A diprotic acid has two protons e.g. H_2SO_4 . As the mol ratio of $NaOH:H_2SO_4$ is 2:1 we need half the mol. The answer is 'a'.
- 10.(2016:10-b) A conjugate acid base pair must differ by only one hydrogen. So a and d are eliminated. Answer c is a redox reaction. The answer is 'b'.
- 11.(2016:11-d) The Brønsted-Lowry model deals with gain and loss of H^+ and uses H_2O and H_3O^+ so answer 'a' is wrong. Answers 'b' and 'c' are not balanced so the answer is 'd'.
- 12.(2016:12-d) The question seeks to clarify the difference between end point and equivalence point so answer 'a' is wrong. Answer 'b' is only right if the stoichiometric ratio is 1:1 and the concentrations are 1:1 or the stoichiometric ratio $\times 1/\text{concentration}$ is a constant. Answer 'c' is wrong because the pH is not 7 when weak acids or bases are titrated against strong ones. The answer is 'd'.
- 13.(2017:01-d) This question combines knowing of what an acid is and knowing the names of a very limited number of strong acids; including nitric, hydrochloric and sulfuric. The answer is 'd'.
- 14.(2017:08-d) The question reinforces the need to write an equation before starting a chemistry problem. The mol ratio is 2:1 so the volumes will be in the same ratio. You should also note the base was in the flask so the answer is 'd'.
- 15.(2018:01-c) Interesting fact that mixing red and green makes yellow. The solution began basic so the initial indicator state would be green and would, at equivalence, be yellow and then turn red in acid solution. The answer is 'c'.
- 16.(2018:03-b) The conductivity of a solution depends on the concentration of ions present. Thus, the acid with the largest K value and the highest concentration will have the most ions and conduct best. The answer is 'b'.
- 17.(2018:08-b) A conjugate pair differ by just one H. So only (ii) and (iii) meet this condition. The answer is 'b'.
- 18.(2018:18-d) Davy considered an acid to have replaceable hydrogen, while Arrhenius said acids produce H^+ in aqueous solution and bases produce OH^- in aqueous solution. Brønsted/Lowry considered acids as proton donors and bases as proton acceptors. The answer is 'd'.
- 19.(2019:01-c) When boric acid is neutralised by sodium hydroxide solution the borate ion will be formed. This ion will undergo hydrolysis and reform boric acid as it is a weak acid. Hydroxide ions will form, the $[OH^-] > [H^+]$ so the pH will be above 7. To find this equivalence point requires an indicator with an end point of 8-10 – the answer is cresolphthalein – the answer is 'c'.
- 20.(2019:08-b) It is important to realise answer (a) is not correct because that might refer to a high concentration of weak acid and similarly strong acids might be dilute and not produce large amounts of hydrogen ions. The answer is 'b'.
- 21.(2019:16-c) Commercial grade $HCl(aq)$ can be purchased at 12 mol L^{-1} . This has a pH of -1.07. The pH scale is designed for common, especially biological, systems and, in unusual instances, extends beyond 0 and 14. The answer is 'c'.
- 22.(2019:21-b) The experiment involves calcium carbonate and calcium hydrogencarbonate. There was no calcium hydroxide present. The answer is 'b'.
- 23.(2019:24-d) In three cases the underlined substance is acting as a Brønsted-Lowry base. The answer is 'd'.
- 24.(2020:02) A monoprotic acid has only one acidic hydrogen while a polyprotic acid has more than one. To be acidic a hydrogen must bond to an oxygen or atom of high electronegativity eg Cl in HCl but not C in CH_3 . The answer is (b).
- 25.(2020:05) i) is correct ii) is correct
iii) is not true – $[H^+] = [OH^-]$ but the ionisation is not complete
iv) is not true – $[H^+] = [OH^-]$
Answer is (a).
- 26.(2020:15) A conjugate acid base pair are two molecules where the second is formed by the first losing an H^+ . The answer is (b).
- 27.(2020:17) Indicators change over a range of pH values so (b) is wrong. If they were strong they would not change colours. They are weak acids or bases that happen to demonstrate a colour change. The answer is (d).
- 28.(2020:18) Stronger means produces more product – the answer is nitrous acid. This is shown by a higher K_a value. Higher pH means produces fewer H^+ ions – the answer is hydrocyanic acid. The answer is (b).

Chapter 3: Oxidation-Reduction

- 1.(2015:01-a) Answer 'a'. Using rules for assigning ON.
- 2.(2015:05-b) Answer 'b'. $E^\circ = +0.59$.
- 3.(2015:08-a) Answer 'a'. No precipitate and no reaction.
- 4.(2015:10-c) Answer 'c'. Nickel and chromium are green but lead will not react.
- 5.(2015:17-b) Answer 'b'. 1.28 V.
- 6.(2016 SP:15-c) In this cell Mn will be oxidised and Ag^+ will be reduced.
 E° values: $Mn + 1.18$
 $Ag^+ + 0.80$
Total $E^\circ = +1.98$. The answer is 'c'.

- 7.(2016 SP:16-d) A fuel cell is a REDOX cell utilising gases. So answer (i) is correct. Answer (ii) is correct. As the main product of hydrogen fuel cells is water vapour it is a low-emission technology. They cannot be recharged by reversing the flow of electricity. The answer is 'd'.
- 8.(2016:13-d) Calculation of oxidation numbers is specifically mentioned in the syllabus. The negative ON is awarded to the element closest to fluorine. So for nitrogen, Step 1: $0 \rightarrow -3$, Step 2: $-3 \rightarrow +2$, Step 3: $+2 \rightarrow +4$, Step 4: $+4 \rightarrow +5$ a change of 1. The answer is 'd'.
- 9.(2016:14-a) Students might see chromate and dichromate and assume it is a redox reaction in part a. It is in fact acid/base. In all other cases the ON of chromium changes. The answer is 'a'.
- 10.(2016:15-c) This question can be solved using the Data Book. Reducing agents get oxidised so we need to look at equations where the substance is losing electrons. Check the E° values of the reactions making sure you have them all as oxidation reactions. The answer is 'c'.
- 11.(2016:16-c) This question can be solved using the Data Book. Find the E° values of each of the half-equations. Add them and find which total is negative – i.e. it is NOT spontaneous. The answer is 'c'.
- 12.(2016:17-a) The element C reduced both A^{2+} and B^{2+} . Thus it is the most powerful reducing agent. A reduced B^{2+} so the answer is 'a'.
- 13.(2017:02-b) An exercise in using the oxidation number rules that you should know. The answer is 'b'.
- 14.(2017:03-b) This question involves the understanding that for a fixed quantity of electrons the valency of an ion will determine how many mol of product will form. The answer is 'b'.
- 15.(2017:04-c) This question appeared in the 2016 examination as well. It recognises that both galvanic and electrolytic cells have external wires, electrodes and redox reactions. The answer is 'c'.
- 16.(2017:09-c) Corrosion is a redox process and this question is essentially asking which of these is not a redox process. Iron(III) chloride dissolving is not redox. The answer is 'c'.
- 17.(2017:10-a) Before wasting time in calculating the E° for every equation perhaps stop and think. In (d) iodine cannot displace chlorine (it is further down the group). In (c) silver ion and bromide ion will not form elements. In (b) gold does not rust we know that. Then using E° check (a). The answer is 'a'.
- 18.(2018:05-a) The question asks for the E° for the reaction $\text{In}^{3+}(\text{aq}) + 3 \text{e}^- \rightarrow \text{In}(\text{s})$. In the question the dichromate reaction was +1.36 V and the total was +1.70 V so the $\text{In} \rightarrow \text{In}^{3+} + 3 \text{e}^-$ reaction must have been +0.34 V. This question asks for the reverse equation so the E° is -0.34 V.
- 19.(2018:06-d) The question is actually asking 'which metals will have a negative E° for the following reaction $\text{In}^{3+}(\text{aq}) + \text{M}(\text{s}) \rightarrow \text{In}(\text{s}) + \text{M}^{x+}(\text{aq})$?' The reaction $\text{M} \rightarrow \text{M}^{x+} + x \text{e}^-$ must have an $E^\circ < +0.34 \text{ V}$ as the $\text{In}^{3+}(\text{aq}) + 3 \text{e}^- \rightarrow \text{In}(\text{s})$ reaction is -0.34 V. $\text{Ni} = +0.24 \text{ V}$, $\text{Sn} = +0.14 \text{ V}$, $\text{Cu} = -0.34 \text{ V}$, $\text{Ag} = -0.80 \text{ V}$. The answer is 'd'.
- 20.(2018:07-d) Oxidation numbers are: $\text{MoO}_2(+4)$, $\text{Mo}_2\text{O}_7^{2-}(+6)$, $\text{HMoO}_4^{2-}(+5)$. The answer is 'd'.
- 21.(2018:14-c) Note Zr^{4+} is not reduced by any of the metals. This means Zr is a very active metal and so is most easily oxidised. The answer is 'c'.
- 22.(2018:15-a) A reducing agent gets oxidised (loses electrons) and forces another substance to be reduced (accept electrons). Rhenium metal fails to reduce any ions and is therefore the weakest. The answer is 'a'.
- 23.(2018:25-a) In all electrochemical cells (galvanic and electrolytic) ions migrate and oxidation occurs at the anode. The answer is 'a'.
- 24.(2019:06-c) This question involved selecting the appropriate half equation using oxygen and reversing the lead equation. The answer is 'c'.
- 25.(2019:10-d) This question could be done quickly by noting the relative activity of Group 17 elements or it could be done slowly by calculating the E° values. The answer is 'd'.
- 26.(2019:17-b) The quickest way to approach this problem is to realise elements have an oxidation number of zero. This eliminates (i) and (iv) – then by calculation the answer is 'b'.
- 27.(2020:01) $\text{Ho}(\text{s})$ has changed oxidation number from 0 to +3 $\therefore$ it has been oxidised so it is a reducing agent. Hydrogen in water has changed from +1 to 0 in $\text{H}_2(\text{g})$ $\therefore$ it has been reduced so it is an oxidising agent. Answer is (c).
- 28.(2020:03) A REDOX definition question – the answer is (b).
- 29.(2020:06) Galvanic cells are spontaneous – they produce energy in the form of electricity. The answer is (d).
- 30.(2020:07) This question is based on the fact concentrated nitric acid can be an oxidising agent as well as being an acid. In this case it is acting as an oxidising agent and copper is oxidised. (i) is true, (ii) is not true (iii) copper is oxidised. The answer is (a).
- 31.(2020:19) A reducing agent reduces other substances – that is, it donates electrons to them. So we are talking about the reverse reactions. The strongest reverse reaction will be the third with E° of +1.34. The answer is (d).
- 32.(2020:20) The sum of the E° values is negative – this cell is not spontaneous – it must be forced. Therefore, it is an electrolytic cell. The answer is (a).

Chapter 4: Organic Chemistry

- 1.(2015:14-a) Answer 'a'. Hydrogen bonding causes more solubility in water than dispersion forces.
- 2.(2015:22-b) Answer 'b'. Dispersion forces < hydrogen bonding < hydrogen bonding and dipole bonding.
- 3.(2015:23-d) Answer 'd'. Double bonds engage in addition polymerisation.
- 4.(2015:24-d) Answer 'd'. Addition is always across the double bond.
- 5.(2015:25-b) Answer 'b'. III cannot accept or donate H bonds.
- 6.(2016 SP:17-c) Secondary alcohols can be oxidised to form ketones. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ is a secondary alcohol. The answer is 'c'.